IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Received September 28, 2019, accepted October 9, 2019, date of publication October 14, 2019, date of current version October 25, 2019.

Digital Object Identifier 10.1109/ACCESS.2019.2947354

Analysis of Skin Marks Through the Use of Intelligent Things

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This work was supported by the Ministry of Science and Higher Education, Poland, under Grant 0080/DIA/2016/45.

ABSTRACT Melanoma is one of the more frequently encountered cancers, which quite often depends on exposure to ultraviolet radiation occurring in solar radiation. Of course, an additional factor is individual immunity and genetics. It is important to discover the possible spread of the cancer, which in this case depends on the observation of the skin, and above all the marks. The quickest possible detection of neoplastic changes can result in a chance to prolong life. In the era of mobility, where everyone has a camera built into the phone and access to the Internet, it is possible to analyze signs by an application that would analyze the skin. In the case of detected exceptions, the application would exchange information with a publicly available database and consult with a dermatologist about the possible need for urgent hispatological examination. In this paper, we propose a model of such a system based on intelligent things along with a detailed description of individual components. As a classification component, a convolutional classifier was proposed, which accepts not only image data, but also numerical one. The analysis of the proposed solution has been tested and discussed due to numerous advantages and disadvantages of such a solution nowadays.

INDEX TERMS Feature extraction, image processing, deep learning, Internet of Things.

I. INTRODUCTION

Skin melanoma is a malignant tumor that develops via pigmented cells. The factors for melanoma development are primarily UV radiation, which occurs in solar radiation [1], [2]. Other factors are signs on the skin or genetics. The development of the disease is unpredictable and hence difficult to treat. When treating this cancer at an advanced stage, the number of possibilities is minimal [3]. At the moment, one of the most promising therapies is immunotherapy, or stimulation of the body to fight cancer cells [4], [5]. In 2018, James P. Allison and Tasuku Honjo were awarded the Nobel Prize in medicine and physiology for their contributions to immunotherapy. It is estimated that these types of therapies will be much more effective and accessible to people in the next decade. At the moment, receiving immunotherapy is dependent on meeting a huge number of criteria, as well as the attending physician. Hence, quite often other methods of fighting cancer cells are used. However, it should be noted that early diagnosis of cancer may result in a much better prognosis.

The associate editor coordinating the review of this manuscript and approving it for publication was Wei Wei.

In order to reduce the risk of getting sick, avoid too much sun in a large amount, use appropriate sunscreens, or avoid solaria. In addition, it is important to observe the marks on the skin, and to analyze them by the following features.

1) Asymmetry – an irregular shape,
2) Borders – uneven boundaries,
3) Color – a different color, or spot color changes,
4) Diameter – a nominal diameter is greater than 0.5 cm,
5) Elevate – a convexity of the markings above the level of the epidermis.

In case of visible changes, one should go to a dermatologist in order to perform a more detailed examination and possible excision of the lesion and its examination in terms of possible disease changes. A very important aspect is the control of birthmarks and in case of possible changes, contact with a dermatologist. The simplest way is to look at the signs, however, it should be noted that these types of activities are not the simplest ones. The reason may be the low noticeability of changes, or even forgetting about this. To this end, it is worth using the current achievements of science, not only to remind us of checking, but to find the smallest changes and, if necessary, quickly make an appointment with a doctor.

Unfortunately, constant attention and control of birthmarks can be problematic in the speed of today's life. Not only

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the lack of time or even forgetting to check can be problematic, but also continuous checking. Especially the second motif may trim to excessive or even obsessive checking of birthmarks, which may cause us not to notice changes due to getting used to the current state. With such a large amount of technology that is present in our lives, it is worth using it to control and analyze birthmarks. Using a smartphone with a good quality camera, we can take a photo on which there may be visible elements that escape the human eye.

A. RELATED WORKS

From a technical point of view, the best thing would be to take a picture of a particular mole, and the program returned interesting information. It is quite a difficult task, because the program should know how to search, what to look for, as well as remove unwanted elements in the image. For this purpose, the most commonly used methods are artificial intelligence. In [6], the authors presented using genetic algorithm for feature selection in skin images. The acquired features should be classified to see if the sample indicates a disease [7]. A particularly important place among classifiers is occupied by artificial neural networks [8]–[11]. In the conducted research, the researchers focused on designing the architecture of a given type of network to adapt it to the verification of samples containing a picture of a nevus. An important element is not only the extraction of features from images or numerical data, but also classification [12]. Again in [13], the idea of creating hybrid classification solutions has been described in detail. As part of the proposed solution, the authors connected the neural network with the optimization algorithm. However, it is very important to pay attention to the problems of modern classification, which is the need for a large training set as well the huge computational and time expenditure needed to train such a classifier [14].

The resulting methods should be included as a component in the system that can be implemented. The analysis of such systems is presented in [15]. Using a smartphone to analyze the image shown in [16]. The increasing development of not only methods but also technology makes the Internet of medical goods within the reach of not large implementation costs. It is worth noting that the communication of various devices means that some matters can be dealt with much faster and more accurately. In particular, obtaining different data from the user can make the diagnosis more accurate. In this work, we proposed a system based on communication of intelligent things. As part of the research, the method of analyzing the obtained data using the artificial intelligence techniques was also modeled. The proposed solution is a modification of the convolution neural network that classically accepts image files as input. In the proposed solution, a data set consisting of a graphic file and data in numerical form are interpreted as one input for such a classifier. This approach is to improve the classification process by increasing incoming data from various medical devices.

This study are divided into the following section:

1) Introduction which gives a motivation and current state of art in the topic of nevus detection and classification.
2) Description of system composed of intelligent things, which describes the operation of the system and the usage model.
3) The next section describes the use of deep learning in the analysis of skin nevi. This section presents the mathematical model of analysis as well the convolution neural network. An alternative approach to the use of such an architecture that simultaneously accepts numerical and graphic data is described.
4) The next section describes used database and conducted experiments. The results were discussed due to comparisons with existing solutions.
5) The last section summarizes the achieved results.

II. SYSTEM OF INTELLIGENT THINGS FOR THE USE OF LIVE PROTECTION

The user, wanting to use the application on his smartphone, will take a picture of the mark that he wants to analyze. Such a photograph should be analyzed not only in terms of features that may indicate a disease, but also in terms of possible changes. Such action assumes that the data has already been analyzed before, which may result in a comparison of the obtained values. However, for this to be possible, the system must record information about all data sent by the user in a certain database. In the case of photos, they can take up a large amount of data, so the best solution is an external server that will maintain such a database. In addition, a given user may have other sensors mounted in the watch or other device. This data can also affect data analysis. An example of this is the analysis of the temperature in which the user is located - prolonged stay, as well as insolation can affect the development of possible diseases.

Another situation is additional tests done by a dermatologist, such as a dermoscope or a doctor's decision about a particular mark. All these elements can retrieve data, analyze them automatically or through an external server, and then decide on an urgent visit to the doctor, or even uncertainty, which can be verified by a doctor. All these activities can affect the effectiveness of the system itself, which can be improved in real time thanks to additional knowledge.

To illustrate such a system, let us assume that the user may own n different devices that are paired with each other. Each device downloads certain data from time to time. There can be no situation that the data will be downloaded and saved between very small periods of time, because it would cause a possible system overload. Especially if the number of people using this type of solution would be large. Data device that retrieves information from the user, sends it to the server, where it is processed. Depending on the type of data entering, algorithms extruding features and information are used. All information is saved, and then the decision block decides about any action. In every situation, when

the user takes a picture or sends information to the server himself, he receives feedback. In the case when the result of the analysis is suspected of the possibility of a disease, this information is sent to the center, which on the basis of the data is sent to the selected doctor. The doctor, on the basis of the results obtained and their visualization, decides on the need to contact a doctor or additional tests.

From a technical point of view, such a system is built from several modules. The first of these is the ability to add additional devices to pair. The application retrieves information and sends it to the server using the Internet connection, where it is analyzed. In the case when it is the first addition of this information, they are added to the database, if subsequent - depending on their amount added, or overwritten (depending on the data limit). In the case of the next data record, it is additionally performed to compare with the existing ones and depending on the decision of the classifier, this data can be sent to the attending physician. The doctor receives a notification requesting to check the results of computer analyzes. The decision of the doctor is recorded in the database and its dependence, a notification is sent about an urgent visit to the patient.

The physician's decision on the analysis is important from the point of view of the action, because more and more data in the database with the correct doctor's mark is used for further overtraining decision components in order to increase the efficiency of the whole system. The visualization of the operation of such a system is presented in Fig. 1. In addition, the individual components in the drawing are described further along with their exact operation.

III. DEEP LEARNING FOR SKIN MARKS ANALYSIS

Assuming that the sample containing the mole will be graphic, some simplification in processing should be taken into account. We assume that there may be other elements in the picture, apart from signs, which is why each object should be considered separately. This is important due to the situation when there are two or more moles next to each other. In both cases, there is a need to extract this particular markings and exterminate them.

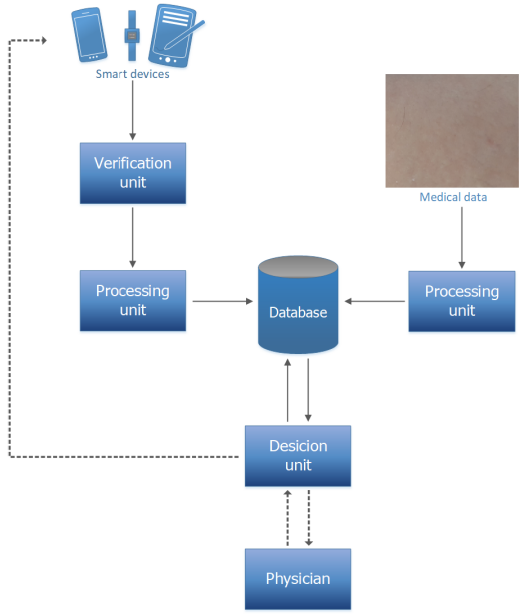


FIGURE 1. Visualization of system operation based on intelligent things. The dashed line means potential action.

Load an image;
Find keypoints using BRISK or SURF;
Create point sets using Eq. (1);
for *each set* **do**
  Create an image by transferring all key points of this set and completing the area of image;
  Resize image;
  Create a set of feature using Eq. (4)–(9);
end
Return images with feature set for further classification using CNN;

Algorithm 1 Image Processing in Verification Unit

A. VERIFICATION UNIT

Data verification consists mainly in the analysis of the image on which various objects can be located, or several signs. The idea of verifying and extracting only the marks has been presented in Alg. 1. On each image, the key points are located, which can be understand as a special configuration of pixels. For this purpose, various algorithms like BRISK [17], or SURF [18] can be used. Using such algorithms we take a certain number of points. Depending on the distance between these points, they are allocated to the appropriate objects. A condition of belonging to a given object is a distance between new key point $\mathbf{x}_{new}$ and any of representative $\mathbf{x}_{rep}$ in this set is smaller or equal radius r, what can be presented as

$$\begin{cases} d\left(\mathbf{x}_{new}, \mathbf{x}_{rep}\right) \leq \dfrac{\sqrt{h \cdot w}}{5}, & \text{add } \mathbf{x}_{new} \text{ to the current object,} \\ d\left(\mathbf{x}_{new}, \mathbf{x}_{rep}\right) > \dfrac{\sqrt{h \cdot w}}{5}, & \text{add } \mathbf{x}_{new} \text{ to a new object,} \end{cases} \tag{1}$$

where $d(\cdot)$ is an Euclidean metric. This selection allows to determine different object on one image. All points in each set gives an area that is cut. As a result, there might be even many new, smaller images (an example of this action can be seen in Fig. 2). All of them are resized to one size and classified using Convolutional Neural Network (CNN) for making a decision if cut out part of the image I is a mark. For this purpose, the CNN is the best classifier. This structure is composed of three type of layer, the first one called convolutional operates on the basis of image filter which is defined as an matrix f. This matrix is moved over the image an changing pixel's value, what can be described as

$$(I * f)(m, n) = \sum_i \sum_j f[i, j] I[m - i, n - j], \tag{2}$$

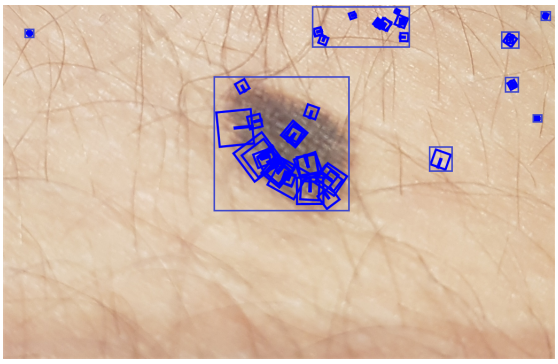


FIGURE 2. Detected key points with assigned objects.

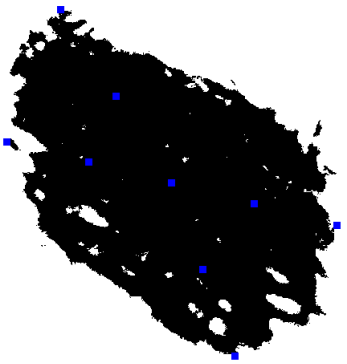


FIGURE 3. A nevus after binarization with the marked position of nine points.

The value in that matrix f are chosen in random way and during the training process, this value are changed. Next type of layer is called pooling, which reduce an image size. This type also uses a matrix, that is moved over the image, but the values in it are taken from image and one of these value is transferred forward. The condition is determined earlier and it can be a maximum, minimum or average function. This two types can appear in large amounts in the network architecture itself. The last type of layer is called fully connected and it a classic neural network. The values of each pixel from the last pooling layer are passed as numerical value to layers of neurons that calculate the activation value of sums of products of weights w_i and values from previous neurons in $t-1$ iteration what can be described as y_i and formulated by

$$y_i^t = a\left(\sum_i w_i \cdot y_i^{t-1}\right), \tag{3}$$

where a function $a(\cdot)$ is an activation.

Adjacent layers are connected to each other by means of synapses, i.e. connections between the smallest elements of the layers – neurons or pixels. Each synapse is burdened with a certain weight, which is modified during training.

In the case of a numerical values in this unit, the verification is based on checking the correct format.

B. PROCESSING UNIT

This unit depends on the input data, in the case of numerical values, they are compared with those in the database and may be rejected depending on the last update. Such a situation will take place when the time interval is too small or the data will be at the same level as the previous entry. Then only information about such a study can be recorded.

In the case of images, the matter is a bit more complicated, because it is necessary to take into account the basic values for a given feature. For this purpose, when the image is not rejected at the level of the verification unit, classic image binarization is performed, i.e. replacing the pixels with black or white to obtain only the contour. It is worth noting that this type of action is performed on a photo sample that has been reduced at an earlier stage – hence the dimension is constant. The following data is downloaded on the modified image

- Asymmetry and borders – this information is gained using simple mathematical formulas. At beginning, a middle point is found using the following formula

$$\mathbf{x}_{middle} = (x, y) = \left(\frac{x_t + x_b}{2}, \frac{y_l + y_r}{2}\right), \tag{4}$$

where coordinates x_t, x_b, y_l and y_r are as follows the highest, lowest point value on the selected axis that represents the mark (after binarization, all white points can be equated with a mark). Then we find the possible radius as

$$r = \frac{x_b - x_t + y_r - y_l}{4}. \tag{5}$$

The values found in this way allow to count the pixels that are in the circle with the center at $\mathbf{x}_{middle}$ and the radius r and defined as γ_1. The values thus obtained will allow to calculate the average quality of asymmetry as θ_1 and borders as θ_2 defined by the following equations

$$\theta_1 = \frac{\gamma_1}{\gamma_2}, \tag{6}$$

$$\theta_2 = \frac{\gamma_3}{\gamma_1}, \tag{7}$$

where γ_2 means all white points and γ_3 is all white points that are not in the circle.

- Color – this value is understood as a possible change color or if there are several colors. Nine pixels are picked using 5 points found in the above feature, additional 4 are points located between the point and the middle one. The visualization of such distribution of points is shown in Fig. 3. Having these points, the average value of the difference is checked, and the other as

$$\xi_i = \frac{1}{3}\sum_{C\in\{R,G,B\}} |C(\mathbf{x}_{middle}) - C(\mathbf{x}_i)|. \tag{8}$$

As a result, we get a set of eight values describing the color of the mark, i.e. $\{\xi_1, \xi_2, \ldots, \xi_8\}$.

- Diameter – the actual size of a nevus is difficult to measure based on the image, hence this value can be given by the user.

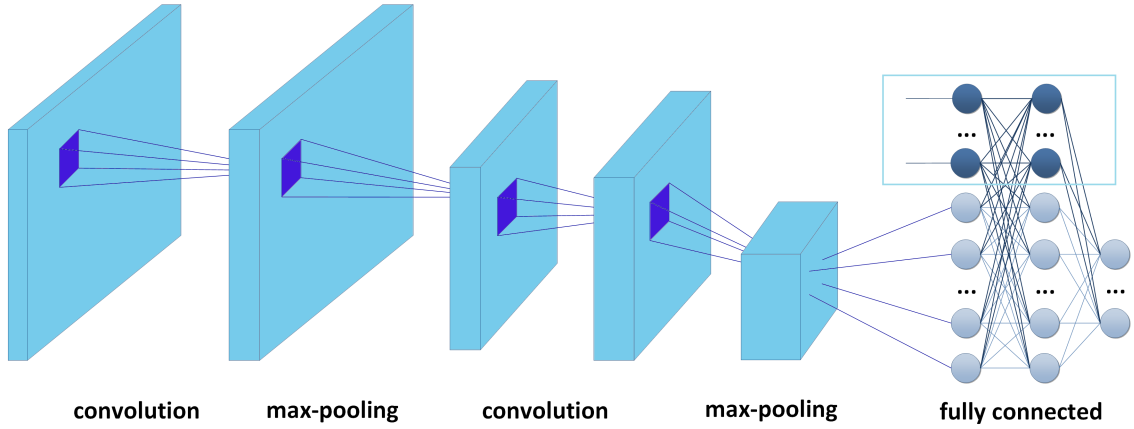


FIGURE 4. The construction of the convolutional neural network, which as input accepts image and numerical data. Numeric data is collected in the fully-connected layer, which is a consequence of additional neurons (darker colors) that adopt these values.

- Elevate - the evolution of the mark is analyzed by correlation of the above elements with respect to the previous assessment, hence in the absence of earlier, this variable μ is equal to 0, and in other case is calculated as

$$\mu = \theta_1 \cdot \theta_2 \cdot \prod_{i=1}^{8} \xi_i. \tag{9}$$

C. DECISION UNIT

When a new record is added or modified, a decision is made on the basis of new information whether the information requires additional doctor's knowledge, or if everything is in order. The effect of such action is returned either to the physician for additional information or for the need to consult the patient or as feedback to the user after entering additional data.

The data is processed by two previous units, i.e. verification and processing, and as a result, the data is used in the classifier, which is to decide on feedback. For this purpose, we propose a modified convolutional network that will simultaneously receive graphic data as well numerical one to combine the features of both types of data contained in the sets. Neural network training consists in adjusting the weights which are initially generated in a random way.

The main modification consists in adding additional neurons in the fully connected layer, which will be responsible for the processing of numerical infromation. For this to be possible, neurons are added to the first column of neurons in this layer, but they are not connected to the previous pooling or convolutional layers. The added neurons are connected to each subsequent neuron in the next columns, but not the previous one. The added units in the first column act as classical input neurons, that is they receive numerical values, they convert them relative to the activation function $a(\cdot)$ and pass forward using synapses. An example of such architecture in the visual form has been shown in Fig. 4.

The training process can take place in one of two types. The first is ignoring added neurons in the fully connected layer and overtraining the classical neural network using only images. If the correct classification level or correct error is obtained, the training should be stopped. Then, selected layers of the network should be frozen, especially the convolutional and pooling layers. Freezing involves not modifying weights in selected layers in the next overtraining. After freezing, the entire fully connected layer, with additional neurons, should be overtrained. In the second overtraining, the image data should be combined with the numerical values.

The second method of training is to use the entire network and complete samples (without division into images and numerical values).

One of the examples of weight modification algorithms is ADaptive Momentum Estimation (Adam) Optimizer [19], which involves the use of classical statistical values such as momentum estimation. The algorithm uses the first two moments, i.e. the mean and variance, which can be described using the following two formulas.

$$m_t = \beta_1 m_{t-1} + (1-\beta_1) g_t, \tag{10}$$
$$v_t = \beta_2 v_{t-1} + (1-\beta_2) g_t^2, \tag{11}$$

where t means the number of iteration, coefficients β_1 and β_2 are constants and g is a gradient value on the current batch. And the estimators of these values have the following form.

$$\hat{m}_t = \frac{m_t}{1-\beta_1^t}, \tag{12}$$
$$\hat{v}_t = \frac{v_t}{1-\beta_2^t}, \tag{13}$$

which are used in the formula of modifying weights in a given iteration $t+1$ using a learning coefficient η and a very small value ϵ in order not to divide by a possible zero.

$$\theta_{t+1} = \theta_t - \hat{m}_t \frac{\eta}{\sqrt{\hat{v}_t} + \epsilon}. \tag{14}$$

IV. EXPERIMENTS

All components of the presented system have been tested and compared in order to achieve the best results. As a part of

TABLE 1. Proposed architecture of convolutional neural network.

Type of layer	Shape
ineine Convolutional 3 × 3	(None,96,96,32)
Activation – ReLu function	(None,96,96,32)
Batch normalization	(None,96,96,32)
MaxPooling 3 × 3	(None,32,32,32)
Dropout 0.25	(None,32,32,32)
Convolutional 3 × 3	(None,32,32,64)
Activation – ReLu function	(None,32,32,64)
Batch normalization	(None,32,32,64)
Convolutional 3 × 3	(None,32,32,64)
Activation – ReLu function	(None,32,32,64)
Batch Normalization	(None,32,32,64)
MaxPooling 2 × 2	(None,16,16,64)
Dropout 0.25	(None,16,16,64)
Convolutional 3 × 3	(None,16,16,128)
Activation – ReLu function	(None,16,16,128)
Batch normalization	(None,16,16,128)
Convolutional 3 × 3	(None,16,16,128)
Activation – ReLu function	(None,16,16,128)
Batch normalization	(None,16,16,128)
MaxPooling 2 × 2	(None,8,8,128)
Dropout 0.25	(None,8,8,128)
Flatten	(None,8192)
Dense 1024	(None,1024)
Activation – ReLu function	(None,1024)
Batch normalization	(None,1024)
Dropout 0.5	(None,1024)
Dense 2	(None,2)
Activation – Softmax function	(None,2)

the tests, the available HAM10000 dataset, ISIC 2018 Lesion Diagnosis Challenge [20], [21] were used. This dataset contains 10015 images arranged in seven following classes.

1) Dermatofibroma – 115 images,
2) Vascular Lesion – 142 images,
3) Actinic Keratosis – 327 images,
4) Basal Cell Carcinoma – 514 images,
5) Benign Keratosis – 1099 images,
6) Melanoma – 1113 images,
7) Nevi – 6705 images.

For test purposes, the dataset was randomly divided in a ratio of 70 : 30 (training : verification), obtaining the following division due to the above classes – 81 : 34, 100 : 42, 229 : 98, 360 : 154, 770 : 329, 780 : 333, and 4694 : 2011.

A. RESULTS FOR THE VERIFICATION UNIT

The analysis started with the selection of the key point search algorithm, which was evaluated in terms of the effectiveness of subsequent verification of the sample whether it is a nevus. In order to select the appropriate architecture of such a network, known configurations such as DenseNet [22], VGG 16 [23], AlexNet [23], Inception [24] and proposed architecture in Tab. 1.

It is worth noting that after using one of the key-point search algorithms – SIFT [25], SURF [18], BRISK [17], or FREAK [26] – points have been assigned to objects. The objects were cut to the points obtained, and the images were scaled relative to the used network architecture. Each of the classifier were trained through 20 iterations using Adam Optimizer. The average effectiveness of the classification has been presented in Tab. 2. This metric is understood as a percentage of correctly classified samples in the database. The best average effectiveness was obtained for the proposed architecture. However, it is worth noting large discrepancies in terms of the algorithm used. The worst results were obtained for the use of the FREAK algorithm, and the best for SURF. For SIFT and SURF, the discrepancy is around 5%, but SURF was made several years after SIFT and in most comparisons it is a more efficient algorithm. Each algorithm analyzes the image in relation to specific features. SURF, which has the best results, uses diffecence of Gaussians. Detailed analyzes using SURF, for each architecture and individual classes are shown on Fig. 5 – 6 and in Tab. 3. The tables presented

TABLE 2. The average effectiveness of the verification unit due to used key-point search algorithm and the chosen structure of the convolutional network.

	DenseNet	VGG 16	AlexNet	Inception	Proposed architecture
ineine SURF	81,44%	74,74%	87,84%	83,91%	88,6%
SIFT	78,11%	74,1%	83,19%	76,21%	81,02%
BRISK	79,23%	76,2%	77,3%	82,6%	81,5%
FREAK	73,22%	72,3%	73,9%	75,2%	75,6%
ine Average	78%	74,34%	80,56%	79,48%	81,68%

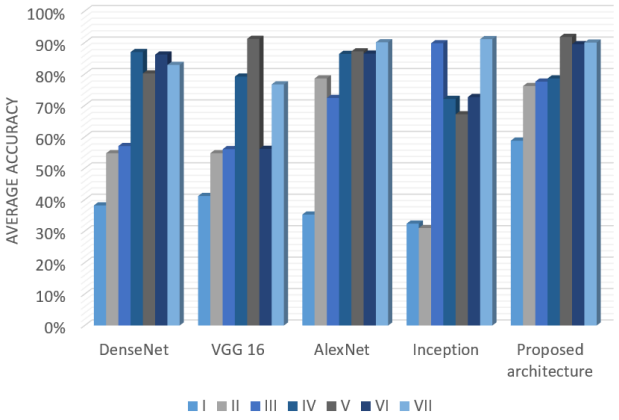


FIGURE 5. A graph of correct verification effectiveness if a given sample contains a nevus for each image class in verification unit.

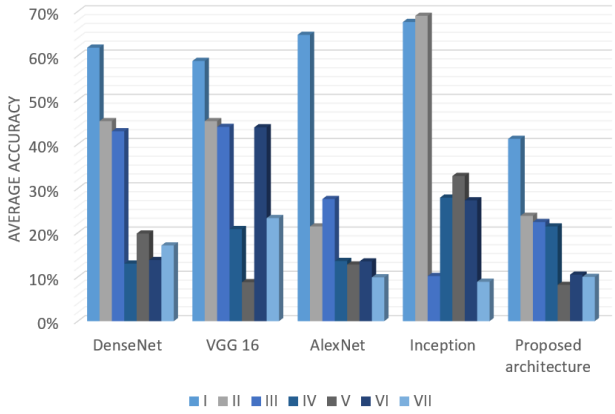


FIGURE 6. A graph of incorrect verification effectiveness if a given sample contains a nevus for each image class in verification unit.

TABLE 3. The average effectiveness of the verification unit due to used key-point search algorithm and the chosen structure of the convolutional network.

DenseNet		VGG16		AlexNet		Inception		Proposed architecture		Class
ine Correctly classified	Incorrectly classified	Correctly classified	Incorrectly classified	Correctly classified	Incorrectly classified	Correctly classified	Incorrectly classified	Correctly classified	Incorrectly classified	
ineine 38,2%	61,8%	41,2%	58,8%	35,3%	64,7%	32,4%	67,6%	58,8%	41,2%	I
54,8%	45,2%	54,8%	45,2%	78,6%	21,4%	31%	69%	76,2%	23,8%	II
57,1%	42,9%	56,1%	43,9%	72,4%	27,6%	89,8%	10,2%	77,6%	22,4%	III
87%	13%	79,2%	20,8%	86,4%	13,6%	72,1%	27,9%	78,6%	21,4%	IV
80,2%	19,8%	91,2%	8,8%	87,2%	12,8%	67,2%	32,8%	91,8%	8,2%	V
86,2%	13,8%	56,2%	43,8	86,5%	13,5%	72,7%	27,3%	89,5%	10,5%	VI
82,9%	17,1%	76,7%	23,3%	90,1%	9,9%	91,1%	8,9%	90%	10%	VII

TABLE 4. The average effectiveness of classifiers with the additional neurons for numerical values to the origin version.

	Original	Proposition
ineine DenseNet	77,30%	79,40%
VGG 16	78,30%	80,01%
AlexNet	77,90%	78,72%
Inception	78,90%	79,25%
Proposed architecture	79,30%	81,40%

present the classification of the test set, which means that the percentage classification of the first category is quite drastic, as none of the classic architectures has reached the threshold 40%, and the proposed one reached almost 20% relative to other classifiers. In other classes, the variation was at a similar level.

B. RESULTS FOR THE DECISION UNIT

The same configurations as above (in the case of verification unit) were used, except that there were three neurons at the exit, which meant decisions such as – no disease, possible changes, cancer. The previously described database was also used with additional numerical descriptions, such as

1) average/minimum/maximum solar radiation,
2) average/minimum/maximum temperature,
3) size marks.

All numerical values were averaged to the interval $\langle 0, 1\rangle$, which caused the addition of additional seven neurons in the fully connected layer. However, the above values have been added in an empirical way to the photo base - only supplemented 30% images in each of the two sets (equally for each) – the rest of them were added with zeros. In addition, each photo was supplemented with decisions depending on the class. Then, selected classifiers' architectures were trained over 30 iterations in both methods described in this paper. The obtained result are presented in Tab. 4. The chart shows a trained classifier for the same base in combination with no proposed modification in this work. For each of the architectures, the efficiency has increased by almost 3%. These results are associated with learning the network from the beginning with full configuration.

The second variant was also used, that is, the classifier was trained in the classic way, only with images samples for 20 iterations. Subsequently, additional neurons were added and blocks were frozen from the very beginning of the architecture. Average efficiency using the block freezing mechanism is shown in Fig. 7. Based on the plotted graph, it is easy to see that freezing one or two blocks does not have a positive effect on effectiveness. However, at 3 or 4 blocks, the effectiveness increases dramatically on average $3 - 4\%$, and with a larger number, the effectiveness decreases slowly. This type of solution achieved a better result than practicing the network from the beginning about $1,2\%$ relative to effectiveness. Unfortunately, the disadvantage is increasing the number of calculations, because weights are modified over 1.5 times more than in the previous case. If the freezing of blocks reduces the amount of modifications needed, adding even one iteration increases the time. And in this case, the classifier needs to overtrain only using images, and then the same images with additional numerical information. Despite these drawbacks, the obtained value is better than other architectures. Also included is the version that the network is not able to make the right decision and this information is forwarded to the doctor. The inclusion of such data, which is uncertain, caused the effectiveness to be lower by an average of 10% for each classifier, which is quite a significant drop.

C. DISCUSSION

The proposed solution, i.e. a combination of extraction of features and a modified convolution network, has shown great potential when performing tests using a publicly available database. The research focused on analysis using various classic architectures of CNN. The obtained results indicate the possibility of increasing the accuracy of classification by using such a classifier, however, with additional neurons and synapses in order to allow classification not only of one type of data, but different. In particular, in the Internet of Things, such solutions have great implementation potential due to the reduction in the number of artificial intelligence methods due to the collected data from various devices.

It is worth noting that additional feature extraction using image processing is also a valuable tool because it causes one image to be processed at two levels – using image processing and a neural classifier. Proposed solution is composed of two units and in both cases, results are better than others approaches. Experiments on verification unit shown that in almost all tested classes, the proposed solution was better. There were some classes, where average classification ratio was smaller than others. The reason for that might be a small

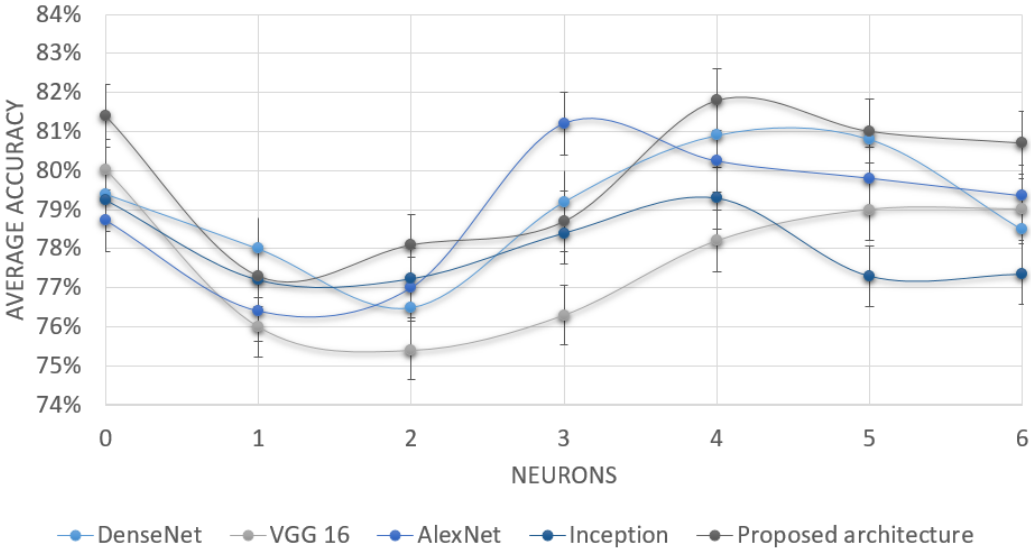


FIGURE 7. Comparison of the average effectiveness of classifiers using the proposed technique of adding numerical neurons.

number of images. Moreover, in conducted tests, we used a four type of key-points algorithm and proved that SURF is much more useful than others for application in every architecture of CNN. In the case of decision unit, obtained results shown that proposed architecture with additional neurons can be a good solution when there are data in different type. Additional neurons for various architectures indicated an increase in correct classification by an average of 3 percent. This result is important due to the potential use of a much larger network architecture.

V. CONCLUSION

The system that allows the addition of additional medical tests is important, and in particular, if the preliminary classification and examination can be performed at any time. This solution is possible using intelligent devices connected with each other to exchange information and quickly process them. This paper proposes a model that is flexible when it comes to adding other elements in a graphical or even numeric form. Trained system, can in a fairly short time return information that will be returned to the doctor. Despite the decrease in efficiency at this stage, the effectiveness of individual components using the public base has shown great practical application at the level of efficiency higher than 80%.

In addition, the work presents a modification of the classic architecture of the convolutional network, which allows for higher efficiency than other popular neural classifiers. It is worth noting that the proprietary architecture allows the addition of various data in two forms, i.e. numeric and graphical data. In future work, we plan to focus on modifying the classifier so that it can receive much larger amounts of data as input. The next element of our research will be to focus on increasing the effectiveness for the classification of birthmarks using data obtaind from other sensors.

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